THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS TWO-CHAPTER TEST SET B

Chapters 13 & 14: **Algebra Play, Area** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The product of two numbers with a fixed sum is largest when the numbers are: 1

(a) one is zero **(b)** equal (or as close as possible)
(c) as different as possible **(d)** both odd

2. The mean of n, $n+2$ and $n+4$ is: 1

(a) $n+2$ **(b)** $n+4$ **(c)** $3n+6$ **(d)** $n+6$

3. In a calendar, the dates in a 2×2 block starting at 20 are 20, 21, 27, 28. Their sum is: 1

(a) 94 **(b)** 98 **(c)** 96 **(d)** 88

4. The expression $(n+1)^2 - n^2$ equals: 1

(a) $n+1$ **(b)** $2n$ **(c)** $2n+1$ **(d)** 1

5. The result of the trick 'Think of n, double it, add 10, halve, subtract n' is: 1

(a) 5 **(b)** 0 **(c)** 10 **(d)** n

6. A rectangle has area 96 cm² and length 12 cm. Its breadth is: 1

(a) 10 cm **(b)** 8 cm **(c)** 7 cm **(d)** 84 cm

7. Which of the following can be the same for two shapes with different areas? 1

(a) colour **(b)** perimeter **(c)** length only **(d)** angles

8. A median of a triangle divides it into two parts of: 1

(a) equal area **(b)** unequal area **(c)** area 1 : 2 **(d)** area 1 : 3

9. The area of a square whose diagonal is 10 cm is: 1

(a) 100 cm² **(b)** 50 cm² **(c)** 20 cm² **(d)** 25 cm²

10. Area of a triangle with sides 3, 4, 5 cm is: 1

(a) 7.5 cm² **(b)** 6 cm² **(c)** 12 cm² **(d)** 10 cm²

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. A trick: Think of n, multiply by 6, add 12, divide by 3, subtract $2n$. What is the answer? 2

12. Express 'the product of two consecutive even numbers' algebraically. 2

13. A triangle has area 54 cm² and base 12 cm. Find its height. 2

14. The diagonal of a rectangle splits it into two triangles. If the rectangle has area 84 cm², find the area of each triangle. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Two numbers add to 37. Find the numbers that give the largest product and the product. 3

16. A triangular field has base 22 m and height 15 m. Find its area and the cost of levelling at ₹ 12 per m². 3

17. Show that 78 and its reverse add to a multiple of 11 and explain algebraically. 3

OR

Show that the difference of a two-digit number and its reverse is a multiple of 9. 3

18. A path 2 m wide runs around a 20 m × 16 m rectangular garden outside it. Find the area of the path. 3

OR

A parallelogram has sides 13 cm and 20 cm and the height on 20 cm is 12 cm. Find its area and the height on the 13 cm side. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. **Case study:** A magician asks: pick a two-digit number, subtract the sum of its digits, and the result is a multiple of 9. 5

(i) Prove it.
(ii) Test with 47.
(iii) Why is it magic?

OR

(i) Find the maximum product of two numbers that sum to 48. 5
(ii) Of three numbers summing to 30 (whole).

20. **Case study:** A kite-shaped garden has diagonals 40 m and 30 m perpendicular to each other. 5

(i) Find its area.
(ii) Compare with a rhombus of the same diagonals.
(iii) Cost of grass at ₹ 8 per m².

OR

(i) Show that the area of a rhombus is $\frac{1}{2}d_1d_2$. 5
(ii) Find the diagonal if the area is 120 cm² and the other diagonal is 15 cm.

END OF QUESTION PAPER — Total: 40 Marks

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